Table A1: Numerical results on Problem 4.1

Init(n)	DFAP-BLS1	DFAP-BLS2	DFAP-BLS3	IHZIPM	SDM-AP	SGM-AP1	SGM-AP2	L-BFGS-AP	MNM-AP
	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$
$x^1(5)$	**11**/8.69e-03/1.51e-07	**11**/3.78e-03/1.51e-07	**11**/1.59e-03/1.51e-07	28/1.23e-02/6.09e-07	24/5.16e-01/3.87e-07	39/4.36e-02/5.93e-07	40/2.55e-02/5.96e-07	–	–
$x^2(5)$	**12**/4.03e-03/4.67e-07	**12**/2.06e-03/4.67e-07	**12**/1.37e-03/4.67e-07	29/9.08e-03/9.30e-07	22/1.73e-02/9.16e-07	65/3.36e-02/7.36e-07	71/3.54e-02/9.21e-07	–	–
$x^3(5)$	**18**/1.28e-03/6.28e-07	**18**/1.48e-03/6.28e-07	**18**/1.20e-03/6.28e-07	28/2.02e-03/6.72e-07	21/1.28e-02/3.93e-07	29/1.44e-02/6.93e-07	29/1.45e-02/6.92e-07	–	–
$x^4(5)$	**10**/2.60e-03/1.51e-07	**10**/2.49e-03/1.51e-07	**10**/2.47e-03/1.51e-07	27/8.30e-03/7.80e-07	20/9.24e-03/9.72e-07	45/1.89e-02/6.80e-07	46/1.81e-02/6.74e-07	–	–
$x^1(5)$	**10**/8.42e-04/7.72e-07	**10**/8.52e-04/7.72e-07	**10**/7.90e-04/7.72e-07	29/9.47e-03/9.49e-07	24/4.57e-02/3.78e-07	43/5.05e-02/6.22e-07	48/5.48e-02/5.82e-07	–	–
$x^2(5)$	**12**/9.38e-04/1.89e-07	**12**/9.86e-04/1.89e-07	**12**/9.44e-04/1.89e-07	29/6.58e-03/8.61e-07	22/1.32e-02/6.82e-07	40/2.30e-02/7.48e-07	51/2.46e-02/8.47e-07	–	–
$x^3(5)$	**14**/1.05e-03/2.06e-07	**14**/1.06e-03/2.06e-07	**14**/9.23e-04/2.06e-07	28/2.37e-03/5.61e-07	20/8.58e-03/9.36e-07	62/2.17e-02/9.15e-07	79/2.65e-02/9.03e-07	–	–
$x^4(5)$	**9**/8.46e-04/7.06e-07	**9**/1.04e-03/7.06e-07	**9**/6.63e-04/7.06e-07	27/1.79e-03/7.76e-07	20/9.54e-03/9.61e-07	28/1.21e-02/5.96e-07	28/1.07e-02/5.93e-07	–	–
$x^1(5)$	**10**/7.30e-04/7.34e-07	**10**/2.09e-03/7.34e-07	**10**/1.41e-03/7.34e-07	30/7.01e-03/7.60e-07	27/1.48e-02/6.07e-07	37/2.35e-02/6.93e-07	81/3.52e-02/6.34e-07	–	–
$x^2(5)$	24/1.78e-03/4.61e-07	24/2.44e-03/4.61e-07	24/1.88e-03/4.61e-07	29/1.04e-02/9.69e-07	**22**/1.21e-02/7.27e-07	36/1.55e-02/7.80e-07	56/2.29e-02/7.77e-07	–	–
$x^3(5)$	**13**/1.38e-03/1.91e-07	**13**/1.05e-03/1.91e-07	**13**/1.11e-03/1.91e-07	28/2.17e-03/5.80e-07	20/9.26e-03/9.74e-07	28/9.54e-03/6.96e-07	29/1.15e-02/6.94e-07	–	–
$x^4(5)$	**10**/7.36e-04/2.27e-07	**10**/7.14e-04/2.27e-07	**10**/7.22e-04/2.27e-07	27/4.69e-03/7.35e-07	20/8.58e-03/9.35e-07	47/1.86e-02/6.39e-07	60/2.14e-02/6.33e-07	–	–

Table A2: Numerical results on Problem 4.2

Init(n)	DFAP-BLS1	DFAP-BLS2	DFAP-BLS3	IHZIPM	SDM-AP	SGM-AP1	SGM-AP2	L-BFGS-AP	MNM-AP
	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$
$x^1(10)$	**169**/1.95e+00/9.20e-07	**169**/1.84e+00/9.20e-07	**169**/1.84e+00/9.20e-07	254/6.82e+00/8.16e-07	–	486/9.03e-01/9.23e-07	486/8.88e-01/9.76e-07	–	–
$x^2(10)$	**116**/1.16e-02/7.54e-07	**116**/1.16e-02/7.54e-07	**116**/1.09e-02/7.54e-07	209/2.36e-02/9.18e-07	270/9.85e-02/9.84e-07	402/1.33e-01/9.53e-07	401/1.31e-01/9.20e-07	–	–
$x^3(10)$	100/9.09e-03/7.57e-07	100/9.53e-03/7.57e-07	100/9.54e-03/7.57e-07	166/1.54e-02/7.55e-07	**59**/2.38e-02/9.63e-07	293/9.36e-02/9.33e-07	297/9.55e-02/9.18e-07	–	–
$x^4(10)$	**107**/9.46e-03/8.05e-07	**107**/9.64e-03/8.05e-07	**107**/9.34e-03/8.05e-07	188/1.60e-02/8.58e-07	137/4.85e-02/9.59e-07	378/1.20e-01/9.94e-07	373/1.24e-01/9.73e-07	–	–
$x^1(10)$	**178**/1.11e+00/9.87e-07	**178**/1.10e+00/9.87e-07	**178**/1.09e+00/9.87e-07	258/5.19e+00/8.54e-07	–	–	–	–	–
$x^2(10)$	**122**/1.09e-02/8.34e-07	**122**/1.07e-02/8.34e-07	**122**/1.18e-02/8.34e-07	206/1.79e-02/8.38e-07	395/1.42e-01/9.87e-07	423/1.33e-01/9.35e-07	424/1.37e-01/9.77e-07	–	–
$x^3(10)$	**119**/1.06e-02/7.67e-07	**119**/1.07e-02/7.67e-07	**119**/1.09e-02/7.67e-07	167/1.67e-02/9.03e-07	288/1.08e-01/9.79e-07	391/1.21e-01/9.81e-07	381/1.22e-01/9.62e-07	–	–
$x^4(10)$	**130**/1.14e-02/7.58e-07	**130**/1.20e-02/7.58e-07	**130**/1.20e-02/7.58e-07	212/1.87e-02/9.65e-07	475/1.66e-01/9.71e-07	453/1.47e-01/9.46e-07	457/1.44e-01/8.70e-07	–	–
$x^1(10)$	**219**/2.44e+00/7.76e-07	**219**/2.46e+00/7.76e-07	**219**/2.45e+00/7.76e-07	343/8.54e+00/8.88e-07	–	–	–	–	–
$x^2(10)$	**196**/1.82e-02/9.32e-07	**196**/1.88e-02/9.32e-07	**196**/1.77e-02/9.32e-07	277/2.47e-02/8.93e-07	–	–	488/1.66e-01/9.56e-07	–	–
$x^3(10)$	**143**/1.30e-02/8.65e-07	**143**/1.27e-02/8.65e-07	**143**/1.32e-02/8.65e-07	234/2.15e-02/8.36e-07	495/1.87e-01/9.87e-07	432/1.85e-01/8.56e-07	425/1.69e-01/9.85e-07	–	–
$x^4(10)$	**183**/1.78e-02/9.62e-07	**183**/1.81e-02/9.62e-07	**183**/1.90e-02/9.62e-07	331/3.15e-02/7.87e-07	–	–	498/1.57e-01/9.19e-07	–	–

Table A3: Numerical results on Problem 4.3

Init(n)	DFAP-BLS1	DFAP-BLS2	DFAP-BLS3	IHZIPM	SDM-AP	SGM-AP1	SGM-AP2	L-BFGS-AP	MNM-AP
	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$
$x^1(1000)$	**19**/5.37e-01/7.07e-07	**19**/5.35e-02/7.07e-07	**19**/6.28e-02/7.07e-07	20/5.73e-01/6.92e-07	31/7.31e-01/5.52e-07	23/4.43e-02/4.84e-07	23/4.47e-02/5.71e-07	31/4.48e-01/7.70e-07	24/2.29e-01/6.84e-07
$x^2(1000)$	**11**/2.26e-02/8.81e-07	**11**/1.64e-02/8.81e-07	**11**/1.66e-02/8.81e-07	16/2.37e-02/8.11e-07	28/5.20e-02/7.68e-07	23/4.18e-02/5.78e-07	22/3.89e-02/9.50e-07	37/4.91e-01/5.67e-07	25/1.00e-01/8.00e-07
$x^3(1000)$	**10**/1.44e-02/7.44e-07	**10**/1.39e-02/7.44e-07	**10**/1.30e-02/7.44e-07	24/3.51e-02/4.78e-07	31/5.04e-02/8.07e-07	21/3.75e-02/7.08e-07	21/3.47e-02/8.07e-07	39/5.19e-01/9.34e-07	21/8.48e-02/9.24e-07
$x^4(1000)$	**12**/1.54e-02/5.55e-07	**12**/1.50e-02/5.55e-07	**12**/1.57e-02/5.55e-07	19/2.87e-02/6.66e-07	32/5.33e-02/9.76e-07	22/3.84e-02/5.92e-07	22/3.57e-02/6.66e-07	39/5.21e-01/9.32e-07	25/9.92e-02/5.90e-07
$x^1(5000)$	18/1.72e-01/9.40e-07	18/1.83e-01/9.40e-07	18/1.75e-01/9.40e-07	**9**/1.50e-01/9.19e-07	31/2.64e-01/6.11e-07	24/1.94e-01/5.54e-07	24/1.98e-01/6.03e-07	38/2.97e+01/5.23e-07	24/9.85e-01/8.81e-07
$x^2(5000)$	11/8.30e-02/7.96e-07	11/7.95e-02/7.96e-07	11/8.34e-02/7.96e-07	**7**/6.16e-02/8.86e-07	28/2.35e-01/8.60e-07	24/2.03e-01/5.29e-07	24/2.02e-01/7.96e-07	30/2.40e+01/5.63e-07	25/9.82e-01/8.95e-07
$x^3(5000)$	**10**/7.92e-02/7.44e-07	**10**/7.98e-02/7.44e-07	**10**/7.64e-02/7.44e-07	24/2.01e-01/4.78e-07	31/2.61e-01/8.07e-07	21/1.78e-01/7.08e-07	21/1.76e-01/8.07e-07	39/3.09e+01/9.34e-07	21/8.20e-01/9.24e-07
$x^4(5000)$	**12**/9.30e-02/5.57e-07	**12**/9.48e-02/5.57e-07	**12**/9.91e-02/5.57e-07	22/1.83e-01/7.99e-07	32/2.75e-01/9.76e-07	22/1.83e-01/5.93e-07	22/1.84e-01/6.67e-07	39/3.30e+01/9.33e-07	24/9.70e-01/6.07e-07
$x^1(10000)$	17/4.15e-01/3.51e-07	17/4.16e-01/3.51e-07	17/4.15e-01/3.51e-07	**7**/3.29e-01/4.59e-07	31/5.16e-01/8.60e-07	24/3.77e-01/7.15e-07	24/3.81e-01/7.67e-07	38/2.41e+02/5.25e-07	24/3.06e+00/6.73e-07
$x^2(10000)$	12/2.02e-01/7.61e-07	12/1.96e-01/7.61e-07	12/1.85e-01/7.61e-07	**8**/1.38e-01/4.17e-08	29/4.82e-01/5.99e-07	24/3.78e-01/5.45e-07	23/3.67e-01/6.73e-07	31/2.05e+02/8.09e-07	24/3.14e+00/9.39e-07
$x^3(10000)$	**10**/1.52e-01/7.44e-07	**10**/1.59e-01/7.44e-07	**10**/1.56e-01/7.44e-07	24/4.12e-01/4.78e-07	31/4.87e-01/8.07e-07	21/3.28e-01/7.08e-07	21/3.44e-01/8.07e-07	39/2.54e+02/9.34e-07	21/2.61e+00/9.24e-07
$x^4(10000)$	**12**/1.94e-01/5.57e-07	**12**/1.83e-01/5.57e-07	**12**/1.91e-01/5.57e-07	23/3.65e-01/8.31e-07	32/5.36e-01/9.76e-07	22/3.58e-01/5.93e-07	22/3.51e-01/6.67e-07	39/2.73e+02/9.33e-07	24/2.96e+00/5.69e-07

Table A4: Numerical results on Problem 4.4

Init(n)	DFAP-BLS1	DFAP-BLS2	DFAP-BLS3	IHZIPM	SDM-AP	SGM-AP1	SGM-AP2	L-BFGS-AP	MNM-AP
	Itr/Tcpu/$\|\psi(x_{\rm final})\|$	Itr/Tcpu/$\|\psi(x_{\rm final})\|$	Itr/Tcpu/$\|\psi(x_{\rm final})\|$	Itr/Tcpu/$\|\psi(x_{\rm final})\|$	Itr/Tcpu/$\|\psi(x_{\rm final})\|$	Itr/Tcpu/$\|\psi(x_{\rm final})\|$	Itr/Tcpu/$\|\psi(x_{\rm final})\|$	Itr/Tcpu/$\|\psi(x_{\rm final})\|$	Itr/Tcpu/$\|\psi(x_{\rm final})\|$
x^1(1000)	**8**/2.83e-02/1.50e-07	**8**/1.60e-02/1.50e-07	**8**/1.72e-02/1.50e-07	11/3.18e-02/3.55e-07	18/4.57e-01/7.94e-07	24/4.90e-02/5.29e-07	24/4.89e-02/6.86e-07	24/3.36e-01/5.29e-07	15/7.39e-02/1.58e-07
x^2(1000)	**6**/1.21e-02/9.91e-07	**6**/1.07e-02/9.91e-07	**6**/8.61e-03/9.91e-07	11/1.85e-02/2.18e-07	19/3.71e-02/5.91e-07	19/3.51e-02/5.46e-07	19/3.76e-02/6.53e-07	19/2.61e-01/5.46e-07	9/4.03e-02/2.36e-07
x^3(1000)	**8**/1.01e-02/6.04e-07	**8**/9.88e-03/6.04e-07	**8**/1.00e-02/6.04e-07	9/1.24e-02/6.62e-07	18/3.29e-02/5.91e-07	18/3.59e-02/9.68e-07	19/3.38e-02/5.84e-07	19/8.85e-01/6.14e-07	13/5.64e-02/1.44e-08
x^4(1000)	**8**/1.01e-02/4.34e-07	**8**/1.02e-02/4.34e-07	**8**/1.04e-02/4.34e-07	11/1.63e-02/5.25e-07	20/3.97e-02/9.87e-07	20/4.13e-02/5.68e-07	20/3.88e-02/7.01e-07	21/2.95e-01/5.12e-07	13/5.35e-02/3.99e-08
x^1(5000)	**8**/5.97e-02/3.45e-07	**8**/6.28e-02/3.45e-07	**8**/6.60e-02/3.45e-07	11/1.05e-01/7.57e-07	19/1.56e-01/8.25e-07	25/2.02e-01/5.91e-07	25/1.99e-01/7.74e-07	25/2.00e+01/5.91e-07	15/5.95e-01/3.52e-07
x^2(5000)	**7**/5.67e-02/2.36e-07	**7**/5.74e-02/2.36e-07	**7**/5.57e-02/2.36e-07	11/1.00e-01/4.76e-07	20/1.68e-01/6.41e-07	20/1.75e-01/6.01e-07	20/1.96e-01/7.26e-07	20/1.66e+01/6.01e-07	9/3.68e-01/5.19e-07
x^3(5000)	**8**/6.21e-02/5.19e-07	**8**/6.12e-02/5.19e-07	**8**/6.09e-02/5.19e-07	9/7.24e-02/6.20e-07	18/1.42e-01/6.32e-07	18/1.44e-01/9.71e-07	19/1.53e-01/5.86e-07	19/2.32e+01/6.15e-07	13/5.10e-01/1.44e-08
x^4(5000)	**8**/6.41e-02/1.12e-07	**8**/6.07e-02/1.12e-07	**8**/6.09e-02/1.12e-07	11/9.33e-02/1.76e-07	21/1.81e-01/5.25e-07	20/1.74e-01/5.70e-07	20/1.72e-01/7.04e-07	21/1.82e+01/5.13e-07	13/5.02e-01/4.00e-08
x^1(10000)	**8**/1.27e-01/4.89e-07	**8**/1.25e-01/4.89e-07	**8**/1.42e-01/4.89e-07	12/2.05e-01/2.15e-07	20/3.32e-01/5.78e-07	25/4.10e-01/8.36e-07	26/4.23e-01/5.53e-07	25/1.70e+02/8.36e-07	15/1.89e+00/4.97e-07
x^2(10000)	**7**/1.17e-01/3.35e-07	**7**/1.16e-01/3.35e-07	**7**/1.16e-01/3.35e-07	11/1.92e-01/6.72e-07	20/3.25e-01/9.02e-07	20/3.82e-01/8.48e-07	21/3.37e-01/5.17e-07	20/1.36e+02/8.48e-07	9/1.17e+00/7.32e-07
x^3(10000)	**8**/1.27e-01/5.10e-07	**8**/1.24e-01/5.10e-07	**8**/1.30e-01/5.10e-07	9/1.57e-01/6.15e-07	18/2.84e-01/6.38e-07	18/2.99e-01/9.71e-07	19/3.09e-01/5.86e-07	19/1.38e+02/6.15e-07	13/1.69e+00/1.44e-08
x^4(10000)	**8**/1.21e-01/1.02e-07	**8**/1.28e-01/1.02e-07	**8**/1.20e-01/1.02e-07	11/1.89e-01/2.08e-07	21/3.33e-01/5.29e-07	20/3.22e-01/5.71e-07	20/3.26e-01/7.05e-07	21/1.51e+02/5.13e-07	13/1.63e+00/4.01e-08

Table A5: Numerical results on Problem 4.5

Init(n)	DFAP-BLS1	DFAP-BLS2	DFAP-BLS3	IHZIPM	SDM-AP	SGM-AP1	SGM-AP2	L-BFGS-AP	MNM-AP
	Itr/Tcpu/$\|\psi(x_{\rm final})\|$	Itr/Tcpu/$\|\psi(x_{\rm final})\|$	Itr/Tcpu/$\|\psi(x_{\rm final})\|$	Itr/Tcpu/$\|\psi(x_{\rm final})\|$	Itr/Tcpu/$\|\psi(x_{\rm final})\|$	Itr/Tcpu/$\|\psi(x_{\rm final})\|$	Itr/Tcpu/$\|\psi(x_{\rm final})\|$	Itr/Tcpu/$\|\psi(x_{\rm final})\|$	Itr/Tcpu/$\|\psi(x_{\rm final})\|$
x^1(1000)	**1**/5.03e-01/0.00e+00	**1**/2.68e-02/0.00e+00	**1**/3.34e-02/0.00e+00	**1**/4.87e-01/0.00e+00	25/4.96e-01/7.63e-07	20/3.86e-02/6.02e-07	6/1.89e-02/1.12e-07	20/2.77e-01/6.02e-07	6/1.02e-01/3.00e-07
x^2(1000)	**1**/8.65e-03/0.00e+00	**1**/1.05e-02/0.00e+00	**1**/7.89e-03/0.00e+00	**1**/2.46e-02/0.00e+00	22/3.67e-02/7.52e-07	22/3.97e-02/7.55e-07	5/1.13e-02/1.97e-08	22/3.01e-01/7.55e-07	5/3.07e-02/2.06e-08
x^3(1000)	**1**/7.58e-03/0.00e+00	**1**/9.25e-03/0.00e+00	**1**/8.73e-03/0.00e+00	**1**/3.02e-02/0.00e+00	20/3.63e-02/5.27e-07	17/3.82e-02/6.95e-07	5/1.62e-02/4.15e-08	19/8.61e-01/8.28e-07	5/3.05e-02/4.90e-08
x^4(1000)	**1**/7.45e-03/0.00e+00	**1**/7.60e-03/0.00e+00	**1**/7.84e-03/0.00e+00	**1**/7.44e-03/0.00e+00	21/3.61e-02/5.35e-07	15/4.04e-02/9.36e-07	5/1.72e-02/4.87e-07	21/2.82e-01/5.45e-07	5/2.78e-02/6.46e-07
x^1(5000)	**1**/4.03e-02/0.00e+00	**1**/4.24e-02/0.00e+00	**1**/4.17e-02/0.00e+00	**1**/4.48e-02/0.00e+00	26/1.99e-01/8.53e-07	21/1.61e-01/6.73e-07	6/4.79e-02/2.51e-07	21/1.63e+01/6.73e-07	6/2.54e-01/6.70e-07
x^2(5000)	**1**/4.20e-02/0.00e+00	**1**/4.06e-02/0.00e+00	**1**/4.26e-02/0.00e+00	**1**/4.12e-02/0.00e+00	23/1.75e-01/8.41e-07	23/1.79e-01/8.44e-07	5/4.01e-02/4.40e-08	23/1.82e+01/8.44e-07	5/2.12e-01/4.60e-08
x^3(5000)	**1**/4.11e-02/0.00e+00	**1**/4.17e-02/0.00e+00	**1**/4.23e-02/0.00e+00	**1**/4.19e-02/0.00e+00	20/1.53e-01/5.27e-07	17/1.77e-01/6.95e-07	5/8.49e-02/4.15e-08	19/2.18e+01/8.28e-07	5/2.52e-01/4.90e-08
x^4(5000)	**1**/4.04e-02/0.00e+00	**1**/4.14e-02/0.00e+00	**1**/4.23e-02/0.00e+00	**1**/4.21e-02/0.00e+00	21/1.59e-01/5.35e-07	15/2.06e-01/9.36e-07	5/8.07e-02/4.87e-07	21/1.80e+01/5.45e-07	5/2.67e-01/6.46e-07
x^1(10000)	**1**/1.60e-01/0.00e+00	**1**/1.65e-01/0.00e+00	**1**/1.58e-01/0.00e+00	**1**/1.60e-01/0.00e+00	27/4.26e-01/6.03e-07	21/3.29e-01/9.52e-07	6/9.19e-02/3.55e-07	21/1.37e+02/9.52e-07	6/7.71e-01/9.48e-07
x^2(10000)	**1**/1.62e-01/0.00e+00	**1**/1.63e-01/0.00e+00	**1**/1.53e-01/0.00e+00	**1**/1.54e-01/0.00e+00	24/3.72e-01/5.95e-07	24/3.67e-01/5.97e-07	5/7.49e-02/6.23e-08	24/1.62e+02/5.97e-07	5/6.56e-01/6.51e-08
x^3(10000)	**1**/1.63e-01/0.00e+00	**1**/1.62e-01/0.00e+00	**1**/1.53e-01/0.00e+00	**1**/1.50e-01/0.00e+00	20/3.48e-01/5.27e-07	17/4.66e-01/6.95e-07	5/2.57e-01/4.15e-08	19/1.36e+02/8.28e-07	5/8.27e-01/4.90e-08
x^4(10000)	**1**/1.66e-01/0.00e+00	**1**/1.59e-01/0.00e+00	**1**/1.55e-01/0.00e+00	**1**/1.54e-01/0.00e+00	21/3.54e-01/5.35e-07	15/5.98e-01/9.37e-07	5/2.57e-01/4.87e-07	21/1.49e+02/5.45e-07	5/8.15e-01/6.46e-07

Table A6: Numerical results on Problem 4.6

Init(n)	DFAP-BLS1	DFAP-BLS2	DFAP-BLS3	IHZIPM	SDM-AP	SGM-AP1	SGM-AP2	L-BFGS-AP	MNM-AP
	Itr/Tcpu/$\|\psi(x_{\rm final})\|$	Itr/Tcpu/$\|\psi(x_{\rm final})\|$	Itr/Tcpu/$\|\psi(x_{\rm final})\|$	Itr/Tcpu/$\|\psi(x_{\rm final})\|$	Itr/Tcpu/$\|\psi(x_{\rm final})\|$	Itr/Tcpu/$\|\psi(x_{\rm final})\|$	Itr/Tcpu/$\|\psi(x_{\rm final})\|$	Itr/Tcpu/$\|\psi(x_{\rm final})\|$	Itr/Tcpu/$\|\psi(x_{\rm final})\|$
x^1(1000)	**5**/5.03e-01/6.66e-07	**5**/3.33e-02/6.66e-07	**5**/3.55e-02/6.66e-07	7/5.02e-01/4.49e-07	26/4.83e-01/8.08e-07	26/6.37e-02/8.10e-07	21/5.06e-02/6.19e-07	26/3.70e-01/8.10e-07	7/1.75e-01/6.76e-08
x^2(1000)	**5**/1.29e-02/6.66e-07	**5**/1.41e-02/6.66e-07	**5**/1.19e-02/6.66e-07	7/3.58e-02/4.49e-07	27/5.05e-02/6.16e-07	27/5.92e-02/6.17e-07	22/4.67e-02/4.76e-07	27/3.73e-01/6.17e-07	8/7.18e-02/5.41e-08
x^3(1000)	**5**/1.25e-02/5.11e-07	**5**/1.16e-02/5.11e-07	**5**/1.18e-02/5.11e-07	7/4.00e-02/4.49e-07	27/5.18e-02/6.39e-07	27/5.76e-02/6.40e-07	21/4.24e-02/9.78e-07	27/3.66e-01/6.40e-07	9/7.36e-02/2.77e-08
x^4(1000)	**5**/1.11e-02/4.83e-07	**5**/1.17e-02/4.83e-07	**5**/1.16e-02/4.83e-07	7/1.72e-02/4.49e-07	27/5.32e-02/6.38e-07	27/5.56e-02/6.39e-07	24/4.36e-02/1.26e-07	27/3.78e-01/6.39e-07	11/1.03e-01/6.19e-07
x^1(5000)	**4**/7.87e-02/3.20e-07	**4**/6.24e-02/3.20e-07	**4**/6.16e-02/3.20e-07	5/9.23e-02/9.86e-07	27/2.25e-01/9.05e-07	27/2.28e-01/9.05e-07	22/1.84e-01/1.37e-08	27/2.10e+01/9.05e-07	8/4.57e-01/7.59e-08
x^2(5000)	**4**/6.22e-02/3.20e-07	**4**/6.23e-02/3.20e-07	**4**/6.25e-02/3.20e-07	5/8.68e-02/9.86e-07	28/2.33e-01/6.90e-07	28/2.31e-01/6.90e-07	20/1.67e-01/8.19e-08	28/2.27e+01/6.90e-07	8/4.49e-01/1.15e-07
x^3(5000)	**4**/6.50e-02/3.22e-07	**4**/6.25e-02/3.22e-07	**4**/6.31e-02/3.22e-07	5/7.52e-02/9.86e-07	28/2.36e-01/7.16e-07	28/2.47e-01/7.16e-07	20/1.87e-01/8.50e-08	28/2.33e+01/7.16e-07	8/4.96e-01/1.19e-07
x^4(5000)	**4**/6.16e-02/3.23e-07	**4**/6.19e-02/3.23e-07	**4**/6.42e-02/3.23e-07	5/7.46e-02/9.86e-07	28/2.49e-01/7.15e-07	28/2.38e-01/7.16e-07	23/2.11e-01/1.09e-08	28/2.41e+01/7.16e-07	11/6.91e-01/1.60e-08
x^1(10000)	**4**/2.51e-01/1.15e-07	**4**/2.07e-01/1.15e-07	**4**/2.01e-01/1.15e-07	5/2.22e-01/3.49e-07	28/4.81e-01/6.40e-07	28/5.56e-01/6.40e-07	20/3.52e-01/7.60e-08	28/1.88e+02/6.40e-07	9/1.59e+00/5.42e-08
x^2(10000)	**4**/2.08e-01/1.15e-07	**4**/2.11e-01/1.15e-07	**4**/2.12e-01/1.15e-07	5/2.40e-01/3.49e-07	28/4.64e-01/9.75e-07	28/4.76e-01/9.75e-07	21/3.51e-01/5.84e-08	28/1.90e+02/9.75e-07	9/1.50e+00/8.18e-08
x^3(10000)	**4**/2.13e-01/1.15e-07	**4**/2.12e-01/1.15e-07	**4**/2.10e-01/1.15e-07	5/2.23e-01/3.49e-07	29/5.12e-01/5.06e-07	29/5.02e-01/5.06e-07	21/3.51e-01/6.07e-08	29/1.98e+02/5.06e-07	10/1.81e+00/4.29e-08
x^4(10000)	**4**/2.10e-01/1.15e-07	**4**/2.14e-01/1.15e-07	**4**/1.98e-01/1.15e-07	5/2.39e-01/3.49e-07	29/5.14e-01/5.06e-07	29/4.90e-01/5.06e-07	22/3.76e-01/3.06e-08	29/2.05e+02/5.06e-07	9/1.69e+00/8.71e-08

Table A7: Numerical results on Problem 4.7

Init(n)	DFAP-BLS1	DFAP-BLS2	DFAP-BLS3	IHZIPM	SDM-AP	SGM-AP1	SGM-AP2	L-BFGS-AP	MNM-AP
	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$
x^1(1000)	**33**/5.94e-02/8.01e-07	**33**/4.46e-02/8.01e-07	**33**/4.07e-02/8.01e-07	52/1.14e-01/8.72e-07	202/8.04e-01/9.15e-07	127/2.46e-01/7.81e-07	115/2.18e-01/8.93e-07	114/1.64e+00/7.19e-07	–
x^2(1000)	**36**/4.41e-02/5.36e-07	**36**/4.71e-02/5.36e-07	**36**/4.68e-02/5.36e-07	57/9.43e-02/9.34e-07	202/3.57e-01/9.16e-07	115/2.20e-01/9.36e-07	111/1.94e-01/9.56e-07	125/2.00e+00/9.78e-07	–
x^3(1000)	**35**/4.23e-02/1.37e-07	**35**/4.26e-02/1.37e-07	**35**/4.23e-02/1.37e-07	62/8.23e-02/7.59e-07	202/3.70e-01/9.16e-07	113/2.05e-01/9.04e-07	112/2.03e-01/9.77e-07	108/1.71e+00/9.27e-07	–
x^4(1000)	**25**/3.12e-02/7.16e-07	**25**/3.23e-02/7.16e-07	**25**/3.01e-02/7.16e-07	59/7.81e-02/7.00e-07	202/3.53e-01/9.16e-07	119/2.11e-01/7.77e-07	116/2.01e-01/9.21e-07	118/1.89e+00/9.62e-07	–
x^1(5000)	**39**/2.93e-01/6.71e-07	**39**/2.94e-01/6.71e-07	**39**/2.93e-01/6.71e-07	51/3.84e-01/8.10e-07	229/1.90e+00/8.79e-07	124/1.08e+00/9.82e-07	129/1.11e+00/9.98e-07	142/1.33e+02/6.74e-07	–
x^2(5000)	**47**/3.50e-01/5.15e-07	**47**/3.54e-01/5.15e-07	**47**/3.43e-01/5.15e-07	58/4.45e-01/6.47e-07	229/1.96e+00/8.79e-07	127/1.12e+00/9.79e-07	131/1.15e+00/9.85e-07	130/1.21e+02/6.17e-07	–
x^3(5000)	**43**/3.10e-01/4.90e-07	**43**/3.17e-01/4.90e-07	**43**/3.19e-01/4.90e-07	59/4.59e-01/9.97e-07	229/1.95e+00/8.79e-07	130/1.12e+00/8.84e-07	126/1.09e+00/9.72e-07	133/1.15e+02/9.85e-07	–
x^4(5000)	**45**/3.32e-01/7.04e-07	**45**/3.41e-01/7.04e-07	**45**/3.39e-01/7.04e-07	62/5.00e-01/2.72e-07	229/1.96e+00/8.79e-07	126/1.09e+00/9.08e-07	126/1.10e+00/8.01e-07	139/1.26e+02/9.69e-07	–
x^1(10000)	**34**/5.48e-01/8.05e-07	**34**/5.39e-01/8.05e-07	**34**/5.58e-01/8.05e-07	64/1.03e+00/5.80e-07	241/3.85e+00/8.30e-07	136/2.26e+00/8.84e-07	127/2.09e+00/8.38e-07	144/9.85e+02/7.74e-07	–
x^2(10000)	**52**/8.40e-01/6.08e-07	**52**/8.25e-01/6.08e-07	**52**/8.46e-01/6.08e-07	58/9.55e-01/3.84e-07	241/3.88e+00/8.30e-07	133/2.20e+00/6.34e-07	133/2.21e+00/7.51e-07	137/9.74e+02/8.07e-07	–
x^3(10000)	**53**/8.50e-01/7.57e-07	**53**/8.67e-01/7.57e-07	**53**/8.94e-01/7.57e-07	57/9.42e-01/9.60e-07	241/3.90e+00/8.30e-07	137/2.30e+00/9.40e-07	127/2.13e+00/8.76e-07	138/9.95e+02/7.08e-07	–
x^4(10000)	**52**/8.50e-01/4.00e-07	**52**/8.43e-01/4.00e-07	**52**/8.46e-01/4.00e-07	61/1.01e+00/8.29e-07	241/3.86e+00/8.30e-07	135/2.23e+00/6.77e-07	131/2.16e+00/7.90e-07	135/9.68e+02/4.32e-07	–

Table A8: Numerical results on Problem 4.8

Init(n)	DFAP-BLS1	DFAP-BLS2	DFAP-BLS3	IHZIPM	SDM-AP	SGM-AP1	SGM-AP2	L-BFGS-AP	MNM-AP
	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$	Itr/Tcpu/$\|\psi(x_{\text{final}})\|$
x^1(1000)	**31**/5.46e-02/2.77e-07	**31**/4.44e-02/2.77e-07	**31**/4.12e-02/2.77e-07	55/1.18e-01/5.63e-07	201/1.07e+00/7.99e-07	80/1.49e-01/7.76e-07	87/1.64e-01/8.50e-07	215/2.88e+00/9.98e-07	–
x^2(1000)	**41**/5.30e-02/7.72e-07	**41**/5.97e-02/7.72e-07	**41**/5.50e-02/7.72e-07	51/7.28e-02/9.25e-07	221/3.61e-01/9.21e-07	86/1.44e-01/9.98e-07	98/1.68e-01/4.97e-07	222/3.28e+00/9.93e-07	–
x^3(1000)	**33**/4.28e-02/3.20e-07	**33**/4.82e-02/3.20e-07	**33**/4.44e-02/3.20e-07	52/6.96e-02/8.99e-07	214/3.51e-01/9.28e-07	87/1.45e-01/8.05e-07	89/1.49e-01/9.80e-07	223/3.25e+00/9.03e-07	–
x^4(1000)	**33**/4.51e-02/8.84e-07	**33**/4.67e-02/8.84e-07	**33**/5.10e-02/8.84e-07	46/7.12e-02/3.95e-08	210/3.59e-01/9.13e-07	90/1.51e-01/8.59e-07	93/1.54e-01/7.08e-07	227/3.34e+00/9.72e-07	–
x^1(5000)	**28**/2.30e-01/8.27e-07	**28**/2.34e-01/8.27e-07	**28**/2.34e-01/8.27e-07	50/4.10e-01/9.08e-07	192/1.58e+00/9.88e-07	84/6.98e-01/6.18e-07	85/7.06e-01/6.28e-07	219/1.87e+02/8.21e-07	–
x^2(5000)	**36**/2.76e-01/6.99e-07	**36**/2.76e-01/6.99e-07	**36**/2.78e-01/6.99e-07	56/4.39e-01/6.23e-07	222/1.86e+00/7.96e-07	87/7.27e-01/8.85e-07	87/7.49e-01/7.90e-07	223/1.99e+02/9.24e-07	–
x^3(5000)	**40**/3.07e-01/6.38e-07	**40**/3.13e-01/6.38e-07	**40**/3.11e-01/6.38e-07	48/4.00e-01/8.70e-07	216/1.85e+00/8.10e-07	90/7.62e-01/8.07e-07	96/8.16e-01/7.04e-07	225/2.02e+02/9.87e-07	–
x^4(5000)	**32**/2.52e-01/5.58e-07	**32**/2.67e-01/5.58e-07	**32**/2.42e-01/5.58e-07	57/4.57e-01/9.38e-07	213/1.78e+00/7.81e-07	86/7.18e-01/8.36e-07	87/7.36e-01/8.22e-07	223/2.03e+02/9.95e-07	–
x^1(10000)	**27**/4.73e-01/3.98e-07	**27**/4.90e-01/3.98e-07	**27**/4.76e-01/3.98e-07	56/9.51e-01/9.24e-07	197/3.20e+00/7.70e-07	77/1.24e+00/8.32e-07	92/1.49e+00/9.82e-07	204/1.40e+03/9.31e-07	–
x^2(10000)	**43**/7.54e-01/3.59e-07	**43**/7.42e-01/3.59e-07	**43**/7.28e-01/3.59e-07	50/8.44e-01/9.69e-07	218/3.55e+00/8.72e-07	97/1.59e+00/8.19e-07	103/1.68e+00/7.98e-07	221/1.54e+03/9.87e-07	–
x^3(10000)	**36**/6.25e-01/3.08e-07	**36**/5.99e-01/3.08e-07	**36**/6.12e-01/3.08e-07	47/8.07e-01/9.19e-07	212/3.40e+00/8.82e-07	85/1.37e+00/6.41e-07	88/1.44e+00/7.63e-07	224/1.51e+03/8.25e-07	–
x^4(10000)	**18**/2.99e-01/6.13e-07	**18**/2.90e-01/6.13e-07	**18**/2.99e-01/6.13e-07	52/8.64e-01/5.30e-07	208/3.36e+00/8.50e-07	92/1.50e+00/7.31e-07	83/1.33e+00/7.22e-07	225/1.53e+03/9.59e-07	–